

# Mantu Shrivastav

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## EDUCATION

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- 2022 – 2026 Bachelor of Engineering (BE) in Civil Engineering  
**Institute of Engineering, Thapathali Campus** | Tribhuvan University, Kathmandu
- 2018 – 2020 Higher Secondary Education  
**Reliance International Academy**
- 2017 – 2018 Secondary Education  
**Manakamana Secondary School**

## PUBLICATIONS

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- [1] Shrestha, R., **Shrivastav, M.\***, & Pal, R. “Evaluating the Accuracy of Equivalent Static vs. Response Spectrum Method for Seismic Analysis of Buildings of Varying Heights.” *International Journal on Engineering Technology (InJET)*. doi:10.3126/injet.v3i1.86983
- [2] **Shrivastav, M.**, Shrestha, R., & Pal, R. “Comparative Seismic Performance Evaluation of RC Dual System Buildings Under NBC 105:2025 and IS 1893 (Part I):2025 Across Varying Heights.” *Journal of Innovations in Engineering Education*, 9(1), 16–26. doi:10.3126/jiee.v9i1.82600
- [3] Pal, R., Shrestha, R., & **Shrivastav, M.** “Comparative Seismic Performance Evaluation of RC Dual System Buildings Under NBC 105:2020 and IS 1893:2016 Across Varying Heights.” *International Journal on Engineering Technology (InJET)*, 3(2), 217–228. doi:10.3126/injet.v3i2.95576
- [4] **Shrivastav, M.**, Shrestha, R., Pal, R., Budhathoki, P., & K.C., R. “A Python-Based Parametric Investigation of Concrete Grade Effects on Axial-Flexural Interaction Capacity of RC Columns: Implications for Typical Building Construction in Nepal.” *International Journal on Engineering Technology and Infrastructure Development (InJET-InDev)*. (Under Review)
- [5] **Shrivastav, M.**, & Shrestha, R. “Fire Assessment in RCC Structures: Safeguarding Nepal’s Built Environment.” *Civil Transcend, Issue (IV)*, ISSN: 3102-0747.
- [6] **Shrivastav, M.**, Shrestha, R., Chaulagain, R., & Marasini, M. “Understanding the Panama Canal Functionality and Global Impact.” *Civil Transcend, Issue (IV)*, ISSN: 3102-0747.

## RESEARCH & PROJECTS

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### **Design and Analysis of 8 Storied High Rise Building in Kathmandu Valley Using Response Spectrum Method** 2025 – 2026

Modeled and analyzed an 8-story high-rise RCC building in ETABS for gravity and seismic loads; designed structural members per NBC and IS codes.

### **Development and Testing of a Physical Model of Tuned Liquid Column Damper (TLCD) for Seismic Vibration Control** 2024 – 2025

Designed and fabricated a physical model of a Tuned Liquid Column Damper (TLCD) to demonstrate its effectiveness in reducing structural vibrations under simulated seismic loads.

### **Development of Working Model of Panama Canal** 2023 – 2024

Designed and fabricated a physical model of the Panama Canal lock system, demonstrating hydraulic principles of water-level management and gravity-driven ship navigation through sequential lock chambers.

### **Machine Learning-Based Compressive Strength Prediction of Concrete** 2025

Developed machine learning models in Python to predict the compressive strength of concrete across variable mix designs and grades. Implemented and compared multiple regression algorithms, training the models on mix design parameters such as water-cement ratio, aggregate proportions, and cement content.

### **Testing of the Liquefaction Potential of Soil Using Machine Learning** 2025

Developed machine learning models in Python to predict soil liquefaction potential under seismic conditions. Utilized geotechnical datasets to train classification algorithms, evaluating model performance for reliable liquefaction susceptibility assessment.

### **Code-Based Tool Generation for Design of Structural Members in RCC Building Using Python** 2025 – 2026

Developed a Python-based automated design tool for reinforced concrete structural members per NBC and IS codes. The program generates reinforcement details, performs capacity checks, and produces design calculation sheets for beams, columns, and slabs.

### **P-M Interaction Curve for Arbitrary Polygonal and Circular Columns Using Python, HTML, and JavaScript** 2025 – 2026

Created an interactive web application combining Python computational backend with HTML/JavaScript frontend to generate axial-moment interaction curves for polygonal RC columns of arbitrary geometry. Enables structural engineers to visualize and verify column capacity under combined loading.

## ACHIEVEMENTS

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- Best Final Year Project Award, Thapathali Graduate Conference, IOE, Thapathali Campus, Tribhuvan University — Project: Structural Analysis of a High-Rise RCC Building Integrated with Code-Based Tools for Member Design and P-M Interaction Curve Generation (*Selected by Department Project Committee*)
- 1st Place, National Level Earthquake Resistant Building Model Competition, Nepal Engineers Association (NEA) — Project: TLCD
- 1st Place, National Level Model Competition, Nepal Engineering College (NEC) — Project: TLCD
- 2nd Place, Sankalpa National Level Model Competition, Kathmandu University (KU) — Project: TLCD
- 3rd Place, Adhar National Level Model Competition, Pulchowk Campus — Project: TLCD
- Public Choice Winner, Intra Model Competition, Thapathali Campus — Project: Panama Canal
- 2nd Place, Intra Model Competition, Thapathali Campus — Project: TLCD
- 2nd Place, All Nepal Civil Model Competition (Event: Yathartha), Thapathali Campus — Project: Eddy Current Damping (*as Mentor*)
- Most Innovative Model Prize, All Nepal Civil Model Competition (Event: Adhar), Pulchowk Campus

## ACADEMIC PERFORMANCE

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|--------------|--------------------------------------------------------------------------------|-----------------|
| 2017–2018    | Secondary Education Examination                                                | GPA: 3.8 / 4.0  |
| 2018–2020    | Higher Secondary Education (+2)                                                | GPA: 3.64 / 4.0 |
| IOE Entrance | Institute of Engineering, Tribhuvan University<br>All Nepal Rank: 499 / 15,000 |                 |

## WORK EXPERIENCE & LEADERSHIP

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### Research Associate | Institute of Himalayan Risk Reduction (IHRR) 2026 – Present

- Working as a Research Associate on “*Seismic Synergy: Learning From Traditional Wisdom To Bring A Resilient Future*”, a project focused on seismic vulnerability assessment and heritage conservation of traditional Newari masonry buildings in the Kathmandu Valley.
- Conducting comparative structural analysis in collaboration with an Indonesian partner team to benchmark seismic vulnerability across traditional building typologies.

### Event Organizer | Yathartha 2026, National Level Shake Table Competition 2026

- Organized the National Level Shake Table Competition during Yathartha 2026, managing logistics, participant coordination, and smooth execution of all rounds.

### Technical Mentor | Yathartha 2026, Seismic Resistant Model Competition 2026

- Mentored the Eddy Current Damping group in the seismic resistant building model competition; guided the team on damping principles and structural design strategies.

### Group Leader | TLCD Physical Model for Seismic Vibration Control 2024 – 2025

- Led a team in designing and fabricating a physical model of a Tuned Liquid Column Damper (TLCD) to demonstrate vibration reduction under simulated seismic loads.
- Coordinated task distribution, managed project timeline, and guided technical decisions throughout the fabrication and testing process.

### Mathematics Teacher (Class 5–8) | Adarsh Secondary School, Gaur, Nepal 2024

- Delivered conceptual and fundamentals-oriented Mathematics lectures to students from Grade 5 to 8.
- Developed problem-solving skills and strengthened mathematical foundations for young learners.

### Group Leader | Working Model of Panama Canal 2023 – 2024

- Led the team in designing and fabricating a physical model of the Panama Canal lock system demonstrating hydraulic water-level management and gravity-driven ship navigation.
- Oversaw planning, material selection, and construction of sequential lock chambers to accurately represent the canal’s operational principles.

### Mathematics Teacher (Class 9–10) | Ananda Bhairab Secondary School, Mulpani 2021

- Delivered conceptual and exam-oriented Mathematics lectures to Grade 9 and 10 students.
- Prepared students for board examinations through targeted problem-solving sessions and practice tests.

- Served as class representative during higher secondary education, facilitating communication between students and faculty.
- Organized academic and co-curricular activities, demonstrating leadership and organizational abilities.

## TECHNICAL SKILLS

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|---------------|----------------------------------------------|
| Programming   | Python, MATLAB, C, C++, VBA, $\text{\LaTeX}$ |
| Structural    | ETABS, SAP 2000, SAFE                        |
| CAD & Survey  | AutoCAD                                      |
| Data Analysis | SmartPLS, Excel                              |
| Office        | Microsoft Word, Advance Excel, PowerPoint    |

## CERTIFICATIONS

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- Machine Learning & Artificial Intelligence with Python — Python for data science, ML model implementation, and AI integration in civil infrastructure.
- Advance Excel, Data Analysis, Graphs and Chart Generation, VBA and Macros
- AutoCAD, ETABS

## SKILLS & ATTRIBUTES

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- Strong communication and interpersonal skills in academic and professional settings.
- Effective team collaborator with proven time management and organizational abilities.
- Creative problem-solver capable of applying engineering principles to novel challenges.
- Able to perform reliably under pressure in both field and competitive environments.

## REFERENCES

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**Dr. Sujan Tripathi**

Assistant Professor  
MSc. Coordinator, Earthquake  
Engg.  
Dept. of Civil Engineering, IOE,  
TU  
Thapathali Campus, Kathmandu  
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**Dr. Alin Chandra Shakya**

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